



CSE-476: Introduction to Natural Language Processing
Quiz 1

Multiple Choice Questions (Single correct)

Instructions: Select the correct option with a short explanation.

- Q1. In the BIO tagging scheme for NER, what does the ``B-" prefix indicate?
- A. Beginning of an entity
 - B. Inside an entity
 - C. Outside any entity
 - D. Between two tokens
- Q2. Which statement best describes a (plain) Markov chain in sequence tagging?
- A. It uses only words (observations) and ignores labels
 - B. It is identical to an HMM
 - C. It models transitions between labels but does not consider the words
 - D. It models emissions $P(\text{word} \mid \text{label})$
- Q3. In a Hidden Markov Model (HMM) for NER, an emission probability is:
- A. $P(\text{label} \mid \text{previous label})$
 - B. $P(\text{label} \mid \text{word})$
 - C. $P(\text{word} \mid \text{previous word})$
 - D. $P(\text{word} \mid \text{label})$
- Q4. Which of the following best describes Semantic Role Labeling (SRL)?:
- A. A task that identifies predicates in a sentence and labels roles of elements such as agent, theme, or location

- B. A process that assigns grammatical roles like subject and object based on syntactic structure
 - C. A method that classifies named entities and links them to a knowledge base
 - D. A technique that labels words with their parts of speech and morphological features
- Q5. Which of the following best describes Byte Pair Encoding (BPE)?
- A. A technique that compresses text by replacing consecutive repeated characters with a count and a symbol
 - B. A tokenization method that iteratively merges the most frequent pair of symbols in a corpus
 - C. A tokenization method that splits text based on whitespace and punctuation
 - D. A character-level encoding scheme that assigns fixed-length binary codes to characters
- Q6. Which of the following best describes the main goal of tokenization in NLP?
- A. To label each token with a grammatical category
 - B. To convert text into a sequence of symbols suitable for modeling
 - C. To reduce vocabulary size to zero
 - D. To assign probabilities to sentences
- Q7. Why is splitting text by spaces (`text.split(" ")`) considered an insufficient tokenizer?
- A. It produces too many tokens
 - B. It does not work well for languages without explicit word boundaries
 - C. It cannot tokenize English at all
 - D. It ignores punctuation entirely
- Q8. Which of the following is NOT listed as a common sequence tagging task?
- A. Part-of-speech tagging
 - B. Named Entity Recognition
 - C. Dependency parsing

D. Sentiment analysis

Q9. What is a common limitation of BIO tagging for Named Entity Recognition?

A. It cannot represent overlapping or nested entities

B. It requires too many labels

C. It does not support multiple entity types

D. It ignores entity boundaries

Q10. When choosing between character-level tokenization and sentence-level tokenization, which trade-off must be considered?

A. Speed vs accuracy

B. Tokenized sequence length vs vocabulary size

C. Precision vs recall

D. Memory vs computation